WINDOWS EVENT LOGS & FINDING EVIL

Forensic Analysis | Technical Learning Portfolio

Sergio Alejandro Perez • September 28, 2026

Hack The Box learning modules | ITDF-2425-MY1 | Instructor: Mr. Valadez

PROJECT OVERVIEW

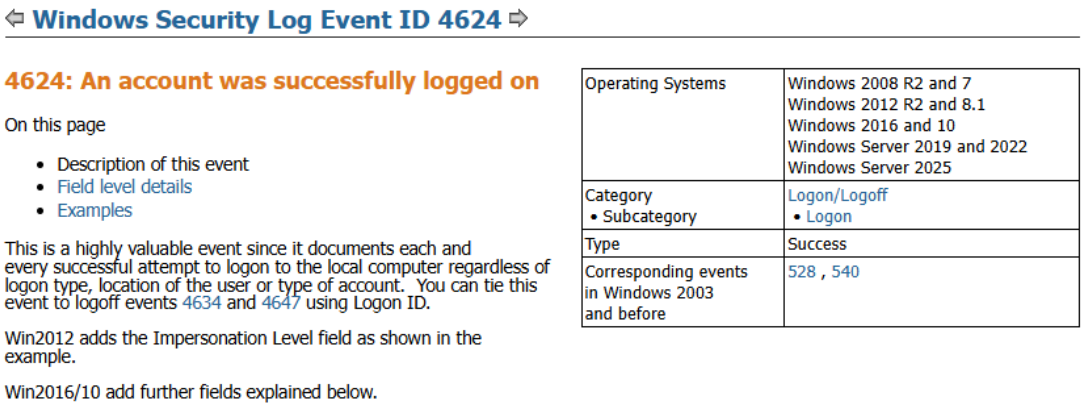
This portfolio documents learning and practical exploration across three Hack The Box sections on Windows Event Logs and threat detection. It covers the Windows Event Viewer, successful-logon event identification, Event Tracing for Windows (ETW), and PowerShell-based event-log discovery.

Focus	Windows log visibility, telemetry, and threat-detection foundations
Tools & concepts	Windows Event Viewer; PowerShell Get-WinEvent; ETW providers
Scope	Coursework-based learning and documented command output; not a live incident investigation

01 Windows Event Logs and Event Viewer

Windows Event Logs record activity generated by the operating system and applications. In the first module, I learned how to locate Event Viewer and recognize its standard Windows Logs categories: Application, Security, Setup, System, and Forwarded Events.

A key example was Security Event ID 4624, which indicates a successful account logon. Reviewing this event can help analysts understand authentication activity and establish a baseline for normal user behavior; an event alone does not prove that activity is benign.



⇦ Windows Security Log Event ID 4624 ⇨

4624: An account was successfully logged on

On this page

- Description of this event
- Field level details
- Examples

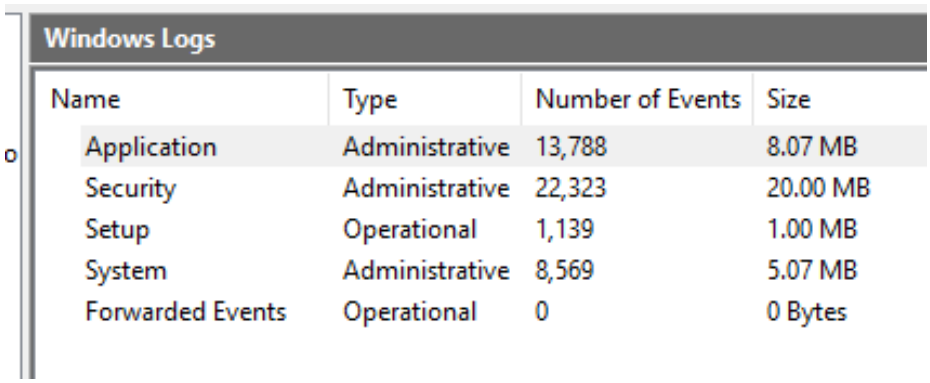
This is a highly valuable event since it documents each and every successful attempt to logon to the local computer regardless of logon type, location of the user or type of account. You can tie this event to logoff events 4634 and 4647 using Logon ID.

Win2012 adds the Impersonation Level field as shown in the example.

Win2016/10 add further fields explained below.

Operating Systems	Windows 2008 R2 and 7 Windows 2012 R2 and 8.1 Windows 2016 and 10 Windows Server 2019 and 2022 Windows Server 2025
Category • Subcategory	Logon/Logoff • Logon
Type	Success
Corresponding events in Windows 2003 and before	528 , 540

Figure 1. Reference material illustrating Windows Security Event ID 4624.



Windows Logs

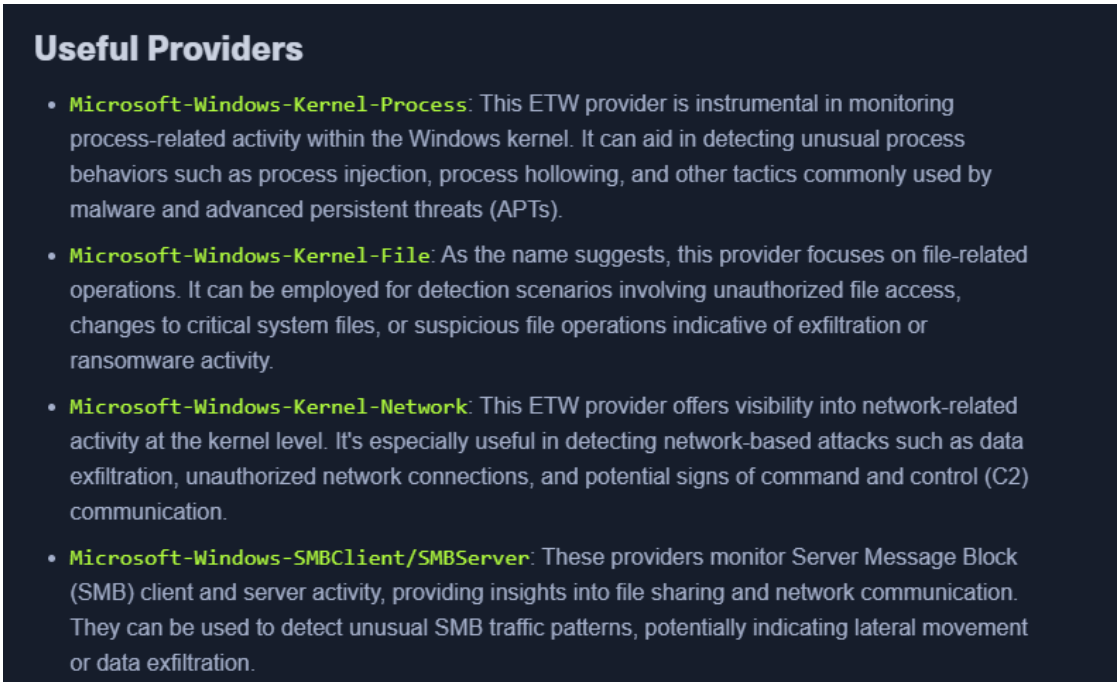
Name	Type	Number of Events	Size
Application	Administrative	13,788	8.07 MB
Security	Administrative	22,323	20.00 MB
Setup	Operational	1,139	1.00 MB
System	Administrative	8,569	5.07 MB
Forwarded Events	Operational	0	0 Bytes

Figure 2. Windows Event Viewer categories and event counts shown in the learning material.

02 Event Tracing for Windows (ETW)

The second module introduced Event Tracing for Windows (ETW), a high-performance Windows tracing framework that can provide telemetry beyond traditional event-log views. I explored how process relationships and other runtime behaviors can offer useful context during threat detection and incident response.

One provider discussed was Microsoft-Windows-DotNETRuntime, which exposes .NET runtime events that may help investigators examine application execution. The material also introduced kernel process, file, and network providers, as well as SMB-related providers, as potential sources of investigative visibility.



Useful Providers

- `Microsoft-Windows-Kernel-Process`: This ETW provider is instrumental in monitoring process-related activity within the Windows kernel. It can aid in detecting unusual process behaviors such as process injection, process hollowing, and other tactics commonly used by malware and advanced persistent threats (APTs).
- `Microsoft-Windows-Kernel-File`: As the name suggests, this provider focuses on file-related operations. It can be employed for detection scenarios involving unauthorized file access, changes to critical system files, or suspicious file operations indicative of exfiltration or ransomware activity.
- `Microsoft-Windows-Kernel-Network`: This ETW provider offers visibility into network-related activity at the kernel level. It's especially useful in detecting network-based attacks such as data exfiltration, unauthorized network connections, and potential signs of command and control (C2) communication.
- `Microsoft-Windows-SMBClient/SMBServer`: These providers monitor Server Message Block (SMB) client and server activity, providing insights into file sharing and network communication. They can be used to detect unusual SMB traffic patterns, potentially indicating lateral movement or data exfiltration.

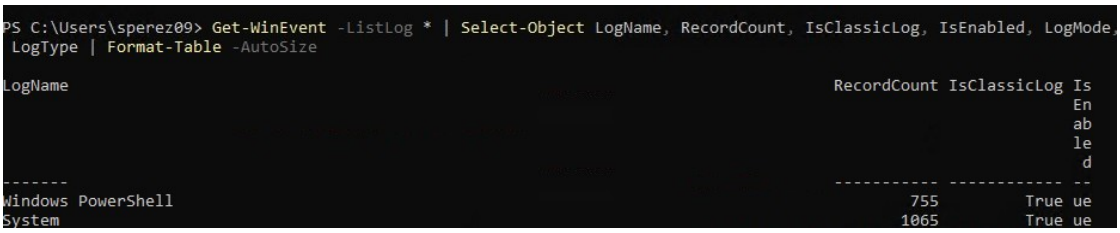
Figure 3. HTB reference slide listing ETW providers and their potential detection uses.

03 PowerShell Log Discovery with Get-WinEvent

The third module focused on reviewing Windows event-log availability at scale using PowerShell. I documented a Get-WinEvent command that enumerates logs and selects properties useful for triage, including log name, record count, log type, and enabled status.

```
Get-WinEvent -ListLog * | Select-Object LogName, RecordCount, IsClassicLog, IsEnabled, LogMode, LogType | Format-Table -AutoSize
```

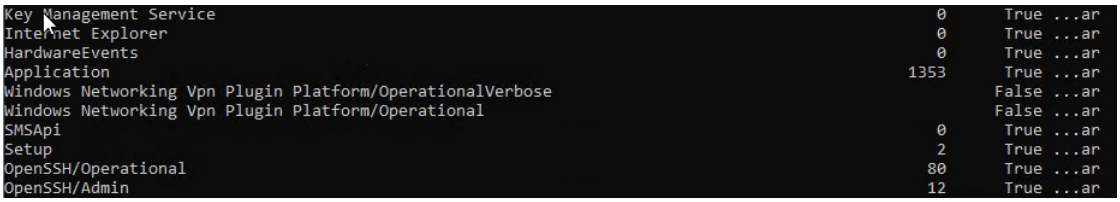
Here, -ListLog * requests available log definitions, the pipe (|) passes output to Select-Object, and Format-Table arranges selected properties for easier reading. The screenshots below document the PowerShell output captured for the assignment.



```
PS C:\Users\sperez09> Get-WinEvent -ListLog * | Select-Object LogName, RecordCount, IsClassicLog, IsEnabled, LogMode,
 LogType | Format-Table -AutoSize

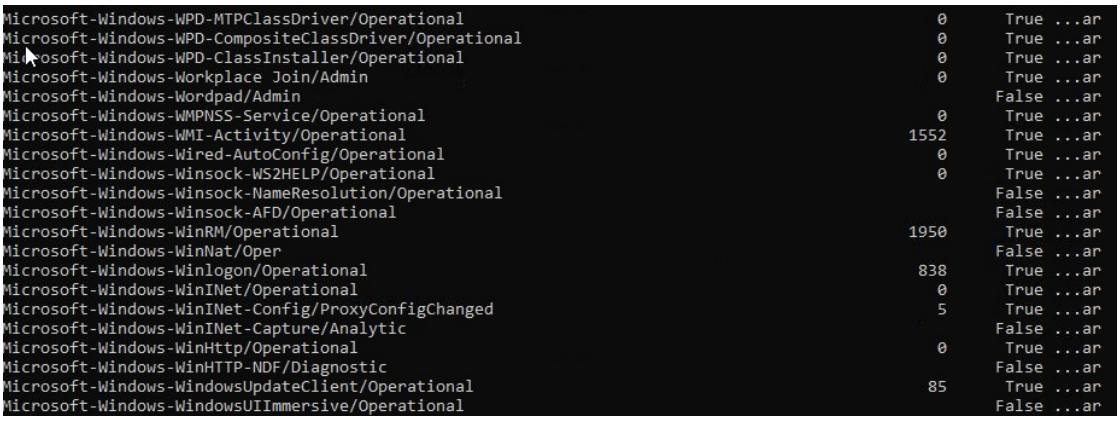
LogName                                                         RecordCount IsClassicLog Is
                                                                                         En
                                                                                         ab
                                                                                         le
                                                                                          d
-------                                                         ----------- ------------ --
Windows PowerShell                                                      755         True ue
System                                                                 1065         True ue
```

Figure 4. PowerShell command and initial event-log listing.



```
Key Management Service                                                    0      True ...ar
Internet Explorer                                                         0      True ...ar
HardwareEvents                                                            0      True ...ar
Application                                                            1353      True ...ar
Windows Networking Vpn Plugin Platform/OperationalVerbose                        False ...ar
Windows Networking Vpn Plugin Platform/Operational                               False ...ar
SMSApi                                                                    0      True ...ar
Setup                                                                     2      True ...ar
OpenSSH/Operational                                                      80      True ...ar
OpenSSH/Admin                                                            12      True ...ar
```

Figure 5. Additional log names and record counts.



```
Microsoft-Windows-WPD-MTPClassDriver/Operational                          0      True ...ar
Microsoft-Windows-WPD-CompositeClassDriver/Operational                    0      True ...ar
Microsoft-Windows-WPD-ClassInstaller/Operational                          0      True ...ar
Microsoft-Windows-Workplace Join/Admin                                    0      True ...ar
Microsoft-Windows-Wordpad/Admin                                                  False ...ar
Microsoft-Windows-WMPNSS-Service/Operational                              0      True ...ar
Microsoft-Windows-WMI-Activity/Operational                             1552      True ...ar
Microsoft-Windows-Wired-AutoConfig/Operational                            0      True ...ar
Microsoft-Windows-Winsock-WS2HELP/Operational                             0      True ...ar
Microsoft-Windows-Winsock-NameResolution/Operational                             False ...ar
Microsoft-Windows-Winsock-AFD/Operational                                        False ...ar
Microsoft-Windows-WinRM/Operational                                    1950      True ...ar
Microsoft-Windows-WinNat/Oper                                                    False ...ar
Microsoft-Windows-Winlogon/Operational                                  838      True ...ar
Microsoft-Windows-WinINet/Operational                                     0      True ...ar
Microsoft-Windows-WinINet-Config/ProxyConfigChanged                       5      True ...ar
Microsoft-Windows-WinINet-Capture/Analytic                                       False ...ar
Microsoft-Windows-WinHttp/Operational                                     0      True ...ar
Microsoft-Windows-WinHTTP-NDF/Diagnostic                                         False ...ar
Microsoft-Windows-WindowsUpdateClient/Operational                        85      True ...ar
Microsoft-Windows-WindowsUIImmersive/Operational                                 False ...ar
```

Figure 6. Continued event-log inventory output.

Skills Demonstrated

- Navigating Windows Event Viewer and identifying standard log categories.
- Interpreting the purpose of successful-logon Event ID 4624.
- Explaining how ETW providers support security visibility and anomaly investigation.
- Using PowerShell Get-WinEvent to enumerate Windows event logs and inspect log metadata.
- Documenting technical learning with supporting screenshots and concise analysis.

Reflection and Next Steps

These modules strengthened my understanding of Windows telemetry as a foundation for digital forensics and threat detection. Event Viewer provides accessible records of system activity, ETW adds additional tracing capabilities, and Get-WinEvent offers a flexible way to inventory logs through PowerShell.

A useful next step would be to build a controlled Windows lab, collect sample authentication events, compare normal and suspicious activity, and document a repeatable investigation workflow with timestamps, filters, and evidence-handling notes.

Project Attribution and Scope

Based on the Hack The Box 'Windows Event Logs & Finding Evil' learning modules and the student's September 2026 coursework submission. Figures include screenshots and reference material from the original assignment. This portfolio summarizes training exercises and learning outcomes; it does not claim a production investigation or confirmed malicious activity.